Olin Scientific and Software Report

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1 Abstract

Olin is an end-to-end scientific software system for gravitational-lensing inverse inference. The system is designed to estimate selected physical parameters of a gravitational-lensing configuration from compatible astronomical observations using a computer-vision model trained on controlled synthetic observations generated from an established physical forward model.

The scientific pipeline begins with a parameterized model of a source and a gravitational lens. Established gravitational-lensing equations are used to generate synthetic observations for which the underlying physical parameters are known exactly. These observations provide controlled training and evaluation data for the machine-learning model. The trained model is then deployed to process compatible real astronomical images supplied by users and estimate the corresponding physical parameters within the defined modelling regime.

The inferred parameters are subsequently passed through the forward physical model to produce a physics-based reconstruction of the observation. This reconstruction provides an additional consistency check on the inference and is included in the generated scientific report together with the predicted parameters, diagnostics, model version, and analysis metadata.

Olin is designed as a complete software workflow rather than solely as a machine-learning experiment. The system provides authenticated image submission, inference, persistent analysis reports, and reproducible model versioning through a web-based interface. The initial system deliberately restricts the physical model and supported observation regime to a defined and controlled scope, allowing the inverse-inference pipeline to be evaluated scientifically before more complex physical effects are introduced.

2 Problem Definition

Gravitational lensing provides a physical mechanism through which the distribution of matter between an observer and a distant astronomical source modifies the apparent position, shape, and brightness of the observed source. The forward problem consists of specifying a physical lens, a source, and the relevant observation geometry, and computing the resulting lensed observation.

The inverse problem considered by Olin is the reverse process. Given a compatible astronomical image containing gravitational-lensing information, the system seeks to estimate a selected set of physical parameters describing the underlying lensing configuration.

Formally, the forward process can be represented as

$$\mathbf{p} \xrightarrow{F} I_{\text{obs}},$$

where $\mathbf{p}$ represents the selected physical parameters of the lensing configuration, F represents the combined physical and observation model, and I_{obs} represents the resulting observed image.

Olin addresses the inverse mapping

$$I_{\text{obs}} \xrightarrow{\hat{F}^{-1}} \hat{\mathbf{p}},$$

where $\hat{\mathbf{p}}$ represents the parameters estimated by the machine-learning model.

The inverse problem is inherently constrained by the assumptions of the forward model. A single two-dimensional astronomical observation does not, in general, uniquely determine an arbitrary three-dimensional physical configuration. Olin therefore does not attempt unrestricted reconstruction of the complete physical state of a gravitational-lensing system. Instead, the system defines a parameterized physical model and estimates only the selected parameters that are identifiable within that model and observation regime.

The primary computational challenge is therefore to learn a robust mapping from observable image structure to physically meaningful parameters while preserving consistency with the underlying gravitational-lensing model.

The system must additionally address the difference between controlled synthetic observations used for supervised learning and real astronomical observations encountered during deployment. This requires explicit evaluation of generalization, observational effects, model assumptions, and the limitations of the resulting inference.

Accordingly, Olin is defined as a physics-grounded inverse-inference system, rather than as a general astronomical image classifier or an unrestricted physical reconstruction system.

3 Scientific Motivation

Gravitational lensing provides a physically grounded inverse problem in which observable image structure contains information about the distribution of matter and the geometry of a distant astronomical system. The relationship between the underlying physical configuration and the resulting observation is described by established gravitational-lensing theory, making the problem well suited to a controlled combination of physical simulation, computer vision, and machine learning.

The central scientific motivation for Olin is to investigate whether a machine-learning model can learn a useful inverse mapping from gravitationally lensed

observations to selected physical parameters when the training data are generated from an explicitly defined physical forward model.

Synthetic data play an important role in this investigation because the underlying physical parameters are known exactly during generation. This allows the inverse model to be evaluated against known ground truth and allows individual physical and observational factors to be varied systematically. Such controlled experiments are difficult to achieve using real astronomical observations alone, for which the complete underlying physical configuration is generally not directly known.

At the same time, the ultimate purpose of the deployed system is not to analyze synthetic images. Olin is intended to accept compatible real astronomical observations and apply the learned inverse model within the defined physical and observational regime. The project therefore treats the transition from controlled synthetic observations to real observations as an explicit scientific validation problem rather than assuming that performance on synthetic data automatically implies real-world validity.

A further motivation is the use of the forward model after inference. By passing the estimated parameters back through the physical model, Olin can produce a physics-based reconstruction of the observation. This provides a second perspective on the inferred solution and allows the system to evaluate whether the predicted parameters produce an observation consistent with the input image.

Olin does not claim to introduce new gravitational-lensing theory or to replace established scientific modelling tools. Its purpose is to construct, evaluate, and deploy a constrained physics-grounded inverse-inference pipeline that connects physical simulation, machine learning, scientific validation, and accessible software delivery.

The project therefore lies at the intersection of gravitational physics, computer vision, machine learning, and scientific software engineering, with the primary emphasis placed on the correctness and evaluation of the central inverse-inference pipeline.

4 Scope and Objectives

4.1 Project Scope

Olin is an authenticated web-based scientific application for performing gravitational-lensing inverse inference on compatible astronomical observations.

The initial system will implement a complete pipeline consisting of:

1. a parameterized gravitational-lensing forward model;
2. controlled synthetic observation generation;
3. supervised training and evaluation of a machine-learning inverse-inference model;

4. evaluation of the model under controlled changes in physical and observational conditions;

5. inference on compatible real astronomical images supplied by users;

6. physics-based reconstruction of an observation from the inferred parameters;

7. generation and persistent storage of a scientific analysis report;

8. authenticated access to submitted analyses and generated reports.

The system will focus on a deliberately constrained physical regime rather than attempting to represent every physical effect present in real gravitational-lensing observations.

4.2 Primary Objectives

The primary objectives of Olin are:

1. To construct a physically interpretable forward model for a defined gravitational-lensing configuration.

2. To generate controlled synthetic observations with known physical parameters for supervised machine-learning experiments.

3. To develop an inverse-inference model capable of estimating a selected set of physical parameters from compatible observations.

4. To evaluate whether the learned model generalizes beyond the exact conditions used during training through controlled test experiments.

5. To evaluate the applicability of the trained model to compatible real astronomical observations.

6. To validate inferred parameters through a physics-based forward reconstruction of the observation.

7. To provide the complete inference workflow through a web application that does not require users to install or execute the underlying machine-learning or simulation software locally.

8. To preserve sufficient analysis and model metadata to make generated results traceable to the model version and configuration used to produce them.

4.3 Scientific Objectives

The scientific component of Olin is intended to answer the following practical question:

> Can a machine-learning model trained on controlled observations generated from an explicitly defined gravitational-lensing forward model recover selected physical parameters from compatible observations with sufficient accuracy and consistency to support a useful analysis workflow?

The project will investigate this question through controlled experiments rather than through a claim of new gravitational-lensing theory.

Scientific evaluation will therefore consider not only parameter prediction accuracy, but also model generalization, observational effects, uncertainty or confidence where supported by the final model, and consistency between the inferred parameters and the corresponding physics-based reconstruction.

4.4 Application Objectives

The deployed application will provide a minimal workflow:

Authenticate → Upload Observation → Analyze → View Results → Generate Report → Retrieve Report

The application will prioritize clarity and reproducibility over a large number of user-facing features.

4.5 Initial Scope Boundaries

The following are outside the initial scope of Olin:

- unrestricted reconstruction of arbitrary three-dimensional mass distributions;
- full relativistic ray-tracing through arbitrary gravitational environments;
- unrestricted multi-plane gravitational lensing;
- modelling every possible telescope, atmospheric, and detector characteristic;
- automatic analysis of arbitrary astronomical images unrelated to the supported gravitational-lensing regime;
- replacement of established scientific gravitational-lensing modelling software;
- generation of new gravitational-lensing theory;

- use of a large language model as part of the core inference pipeline;
- retrieval-augmented generation or document-based knowledge retrieval;
- unnecessary distributed services or infrastructure;
- a large dashboard, analytics platform, or social/profile system.

These exclusions are deliberate scope constraints and may only be changed through an explicit revision of the system specification.

4.6 V1 Physical Regime

The initial physical model will assume a single dominant gravitational lens under the thin-lens approximation. Additional line-of-sight structures and multi-plane lensing are considered future extensions rather than requirements of V1.

The exact lens mass profile, source model, observation model, inferred parameter vector, parameter ranges, and machine-learning architecture will be specified in the corresponding scientific and machine-learning sections of this document before implementation begins.

4.7 Success Criteria

Olin V1 will be considered successful when the complete system can:

1. generate physically defined synthetic observations with known parameters;
2. train and evaluate an inverse-inference model on those observations;
3. demonstrate quantitative performance on held-out synthetic data;
4. demonstrate controlled generalization tests beyond the exact training conditions;
5. process compatible real astronomical observations through the deployed inference pipeline;
6. produce a physics-based reconstruction from the inferred parameters;
7. generate a reproducible scientific report containing the inference result and relevant model metadata; and
8. enforce authentication and ownership of persisted analyses and reports.

Success will not be defined solely by the existence of a working web interface. The scientific validity and measurable behaviour of the inverse-inference model are primary requirements of the project.

5 Physical Foundations

5.1 Observer–Lens–Source Geometry

Gravitational lensing occurs when light from a distant astronomical source passes through the gravitational field of a massive object before reaching the observer. The mass is referred to as the gravitational lens, while the distant object producing the observed light is referred to as the source.

The basic geometry can be represented as

$$\text{Observer} \longrightarrow \text{Lens} \longrightarrow \text{Source}.$$

The observer does not directly see the source in its original position. Instead, the gravitational field of the lens changes the path followed by the light, causing the source to appear at one or more different positions on the sky.

Three distances are important in the lensing geometry:

- D_L: distance from the observer to the lens;
- D_S: distance from the observer to the source;
- D_{LS}: distance from the lens to the source.

In a cosmological treatment these quantities are represented by appropriate angular-diameter distances rather than simple physical distances. Olin will use the distance representation required by the selected lensing model.

The positions of the source and its observed image are represented by angular coordinates rather than ordinary physical coordinates. The unlensed source position is denoted by $\boldsymbol{\beta}$, while the observed image position is denoted by $\boldsymbol{\theta}$.

This distinction is important because an astronomical image records the direction of incoming light on the observer's sky. The image is therefore a two-dimensional projection of a physical system that exists in three dimensions.

Olin does not attempt to reconstruct the complete three-dimensional configuration of the source, lens, and surrounding universe. Instead, the relevant three-dimensional geometry is represented through a defined physical model and its corresponding projected quantities.

The fundamental geometric relationship used throughout the forward model is therefore:

physical source and lens configuration → projected lensing geometry → two-dimensional observation

This establishes the geometric basis for the subsequent gravitational deflection and lens-equation sections.

5.2 Gravitational Deflection

Light follows curved paths when propagating through curved spacetime. In gravitational lensing, the gravitational field of a massive object changes the direction of light travelling from a distant source toward the observer.

For a simple point-mass lens, the magnitude of the physical deflection angle is given by

$$\boxed{\hat{\alpha}(b) = \frac{4GM}{c^2 b}}$$

where G is the gravitational constant, M is the lens mass, c is the speed of light, and b is the impact parameter, representing the perpendicular distance between the light path and the centre of the lens.

The equation captures two important physical behaviours:

- increasing the lens mass increases the amount of deflection;
- decreasing the distance of the light path from the lens centre increases the amount of deflection.

Thus, light passing closer to a massive lens is deflected more strongly than light passing farther away.

The deflection angle $\hat{\alpha}$ describes the physical bending of the light ray near the lens. In the thin-lens formulation, this physical deflection is converted into an angular deflection on the observer's sky by the geometry of the observer, lens, and source.

The reduced deflection is therefore defined as

$$\boxed{\boldsymbol{\alpha} = \frac{D_{LS}}{D_S}\hat{\boldsymbol{\alpha}}}$$

where D_{LS} and D_S describe the lens–source and observer–source geometry respectively.

The distinction between $\hat{\boldsymbol{\alpha}}$ and $\boldsymbol{\alpha}$ is maintained throughout the Olin specification: $\hat{\boldsymbol{\alpha}}$ represents the physical bending of a light ray, while $\boldsymbol{\alpha}$ represents its geometrically scaled effect in the lens equation.

For an extended lens rather than a point mass, the same physical idea applies, but the deflection is produced collectively by the distributed mass of the lens. The corresponding deflection becomes a position-dependent vector field rather than a single value.

This distinction provides the transition from the simple point-mass model used to introduce gravitational deflection to the distributed mass models required for the Olin forward simulation.

5.3 Lens Equation

The lens equation connects the position of a source in the absence of lensing with the apparent position of its lensed image. It provides the central geometric relationship used to describe gravitational lensing.

In vector form, the thin-lens equation is

$$\boxed{\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})}$$

where $\boldsymbol{\beta}$ is the unlensed source position, $\boldsymbol{\theta}$ is the observed image position, and $\boldsymbol{\alpha}$ is the reduced gravitational deflection.

The equation can be understood intuitively as follows: the apparent image position differs from the original source position because the light reaching the observer has been deflected by the gravitational lens.

For a point-mass lens, the equation can be written in terms of the Einstein angle θ_E as

$$\boxed{\beta = \theta - \frac{\theta_E^2}{\theta}}$$

in the corresponding one-dimensional radial representation.

The lens equation may have multiple valid solutions for $\boldsymbol{\theta}$ for a single source position. These solutions correspond to different light paths from the same source reaching the observer and therefore produce multiple observed images.

For Olin, the lens equation forms the conceptual bridge between the physical mass distribution of the lens and the observable structure that appears in the astronomical image.

5.4 Einstein Radius

The Einstein radius is the characteristic angular scale of gravitational lensing for a given lens mass and observer–lens–source geometry. It represents the angular position at which an Einstein ring forms when the source, lens, and observer are perfectly aligned.

For a point-mass lens, the Einstein angle is

$$\boxed{\theta_E = \sqrt{\frac{4GM}{c^2}\frac{D_{LS}}{D_L D_S}}}$$

where M is the lens mass and D_L, D_S, and D_{LS} describe the relevant lensing geometry.

The equation shows that the characteristic angular scale depends on both the mass of the lens and the relative geometry of the observer, lens, and source.

When the source is not perfectly aligned with the lens, the Einstein ring generally breaks into separate images or distorted structures. The Einstein radius therefore provides an important physical scale for understanding the size and structure of lensing features.

In Olin, the Einstein radius is used as a physical reference quantity rather than being treated as an arbitrary image-size parameter.

5.5 Multiple Images and Magnification

A gravitational lens can allow light from the same source to reach the observer along multiple valid paths. As a result, a single background source may appear as multiple images at different positions on the sky.

For the point-mass lens, the lens equation

$$\beta = \theta - \frac{\theta_E^2}{\theta}$$

leads to two image positions for a non-zero source offset:

$$\boxed{\theta_\pm = \frac{\beta \pm \sqrt{\beta^2 + 4\theta_E^2}}{2}}$$

These solutions represent different apparent images produced by different light paths from the same source.

Gravitational lensing also changes the apparent angular area of the source. The corresponding magnification can be expressed as

$$\boxed{\mu = \frac{1}{|\det J|}}$$

where

$$J = \frac{\partial \boldsymbol{\beta}}{\partial \boldsymbol{\theta}}$$

is the Jacobian of the lens mapping.

Intuitively, the Jacobian describes how a small region of the source is stretched or compressed by the lens. A larger apparent area corresponds to magnification, while strong distortions can produce elongated arcs or rings.

Gravitational lensing preserves surface brightness in the ideal geometrical-optics limit, but the apparent area of the source changes. Consequently, the total observed flux of an image can change through magnification.

These effects are important for Olin because the system is intended to work with two-dimensional image structure rather than only with the positions of individual point-like images. The forward model must therefore represent changes in image position, shape, and brightness produced by the lensing mapping.

5.6 Extended Mass Distribution and Thin-Lens Approximation

A real gravitational lens, such as a galaxy, contains matter distributed throughout a three-dimensional volume rather than concentrated at a single point. Its mass distribution can be represented by a volume density

$$\rho(x, y, z).$$

For gravitational-lensing calculations, the mass distribution along the observer's line of sight can be projected onto a two-dimensional lens plane. This produces the surface mass density

$$\boxed{\Sigma(x, y) = \int \rho(x, y, z)\, dz}$$

which represents how much mass is present per unit projected area.

The thin-lens approximation treats this projected mass distribution as being concentrated on a single effective lens plane. This does not mean that the physical lens is actually flat. Rather, it provides a useful approximation in which the three-dimensional mass distribution is represented by its projected effect on passing light.

The projected mass distribution produces a position-dependent deflection field,

$$\boxed{\boldsymbol{\alpha} = \boldsymbol{\alpha}(x, y)}$$

because different light rays pass through different regions of the lens.

The resulting conceptual chain is therefore

$$\boxed{\rho(x, y, z) \rightarrow \Sigma(x, y) \rightarrow \boldsymbol{\alpha}(x, y) \rightarrow \text{lensed image}}$$

This formulation allows Olin to represent a physically distributed lens while retaining a computationally manageable two-dimensional lensing model. The exact parameterization of $\Sigma(x, y)$ will be defined as part of the forward physical model.

6 Forward Physical Model

The forward model defines how Olin will generate an observable astronomical image from a specified physical lensing configuration. It connects the physical parameters of the source, lens, and observation geometry to the final image presented to the machine-learning system.

The forward process is conceptually

$$\boxed{\text{Source} + \text{Lens} + \text{Geometry} \rightarrow \text{Lensed Image} \rightarrow \text{Observed Image}}$$

The model will be implemented as a controlled simulation rather than as a complete representation of every physical process present in real astronomical observations.

6.1 Source Model

The source represents the background astronomical object whose light is being lensed. Its intrinsic brightness distribution will be defined by a parameterized two-dimensional model.

The source model must provide sufficient variation in position, size, orientation, and brightness to prevent the machine-learning model from simply memorizing a single fixed source appearance.

The exact source profile and parameter ranges will be selected during implementation based on the desired physical realism and computational cost.

6.2 Lens Model

The lens represents the projected mass distribution responsible for deflecting the source light. Olin will use a parameterized extended mass model rather than treating the lens as an isolated point mass.

The lens model will define a surface mass distribution

$$\Sigma(x, y)$$

from which the corresponding deflection field is obtained. The exact mass profile and its free parameters will be fixed before dataset generation.

6.3 Lensing Transformation

The source brightness distribution is transformed according to the gravitational-lensing equation

$$\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta}).$$

For each position in the image plane, the lens model determines the corresponding source-plane position. The source brightness at that position is then mapped into the lensed image.

This produces the idealized lensed observation,

$$I_{\text{lensed}}(x, y),$$

which contains the characteristic positional distortions, multiple images, arcs, or rings produced by the selected lens configuration.

6.4 Telescope and Detector Model

The ideal lensed image is not directly equivalent to a real astronomical measurement. The observation process introduces effects such as finite spatial resolution, optical blurring, pixelization, and detector noise.

Olin will therefore include a simplified observation layer after the lensing transformation. A general representation is

$$I_{\text{obs}} = I_{\text{lensed}} * \text{PSF} + N,$$

where PSF represents the effective point-spread function and N represents observational noise.

The initial model will use only the observational effects necessary to create a controlled but realistic training distribution. Detailed telescope-specific modelling is outside the initial scope.

6.5 Noise Model

Noise will be treated as part of the observation process rather than as part of the gravitational-lensing physics.

The noise model will introduce controlled variation between the ideal physical prediction and the final simulated observation. Its parameters will be varied during dataset generation so that the trained model is not restricted to a single noise condition.

The exact noise distributions and parameter ranges will be determined during implementation and evaluation.

6.6 Complete Forward Pipeline

The complete forward simulation can therefore be summarized as

$$\boxed{\mathbf{p} \rightarrow \text{Source} + \text{Lens} \rightarrow I_{\text{lensed}} \rightarrow \text{Observation Model} \rightarrow I_{\text{obs}}}$$

where $\mathbf{p}$ contains the selected physical parameters of the simulated system.

The same forward model will later be used to reconstruct an observation from the parameters predicted by the machine-learning model. This ensures that training data generation and physics-based validation are based on the same defined physical assumptions.

7 Inverse-Inference Problem

The inverse-inference stage of Olin attempts to recover selected physical parameters of a gravitational-lensing configuration from an observed image.

The forward model produces an observation from known parameters:

$$\mathbf{p} \rightarrow I_{\text{obs}}.$$

The machine-learning model performs the reverse mapping:

$$\boxed{I_{\text{obs}} \rightarrow \hat{\mathbf{p}}}$$

where $\hat{\mathbf{p}}$ represents the parameters predicted by the model.

7.1 Inference Target

Olin will estimate only a selected subset of the physical parameters defined by the forward model. The target parameter vector will be chosen such that the parameters are sufficiently identifiable from the available image information.

The exact parameter vector, parameter ranges, and output representation will be fixed after the forward model and initial simulation experiments are completed.

7.2 Training Objective

Synthetic observations provide the required ground truth because the parameters used to generate each image are known exactly. The model can therefore be trained by comparing its predictions with the known parameters.

For a parameter vector $\mathbf{p}$ and prediction $\hat{\mathbf{p}}$, a basic regression objective is

$$\boxed{L_{\text{parameter}} = \|\hat{\mathbf{p}} - \mathbf{p}\|^2}$$

with the exact loss and parameter-wise weighting determined during model development.

7.3 Physics-Based Validation

Parameter accuracy alone is not sufficient to determine whether the inferred solution is physically consistent. The predicted parameters can therefore be passed through the same forward model used to generate the observations:

$$\hat{\mathbf{p}} \rightarrow F \rightarrow \hat{I}_{\text{reconstructed}}.$$

The reconstructed image can then be compared with the input observation to evaluate whether the predicted physical configuration produces a compatible lensing pattern.

This provides an additional validation path:

$$\boxed{I_{\text{obs}} \rightarrow \hat{\mathbf{p}} \rightarrow \hat{I}_{\text{reconstructed}}}$$

The reconstruction is therefore treated as a physics-based consistency check, not as a replacement for the original observation.

7.4 Generalization

The inverse model will be evaluated on observations that differ from the training distribution within the defined physical regime. Tests may vary parameters such as source properties, lens properties, noise, or observational conditions.

The purpose is to determine whether the model has learned meaningful relationships between image structure and physical parameters rather than simply memorizing the appearance of the training data.

The final inference problem is therefore defined by three requirements:

1. accurate prediction of the selected physical parameters;
2. consistency of those predictions with the forward physical model;
3. robustness to controlled variation within the supported observation regime.

8 Synthetic Dataset

The synthetic dataset will provide the controlled training and evaluation data for Olin's inverse-inference model. Each sample will be generated by the forward physical model using known physical parameters.

The dataset generation process is

$$\boxed{\text{Sample Parameters} \rightarrow \text{Forward Model} \rightarrow \text{Synthetic Observation} + \text{Ground Truth}}$$

where the ground truth consists of the parameters used to generate the observation.

8.1 Parameter Variation

The dataset will vary the physical and observational parameters defined by the forward model within specified ranges. This may include source properties, lens properties, source position, observational resolution, and noise conditions.

Parameter ranges will be selected to cover the intended inference regime without generating physically unsupported or unnecessarily difficult samples.

8.2 Dataset Splitting

The dataset will be divided into training, validation, and test sets.

The test set will contain samples that are not used during training or model selection. Where appropriate, parameter ranges or combinations will also be held out to evaluate whether the model generalizes beyond observations that closely resemble its training samples.

Synthetic samples will be generated with reproducible parameter configurations and dataset-generation settings so that experiments can be repeated and compared across model versions.

8.3 Controlled Difficulty

The dataset will include controlled variation in observational conditions rather than assuming that every synthetic observation is ideal.

Noise, resolution, source variation, and other selected observational effects will be introduced systematically. This allows model performance to be measured under progressively different conditions and helps identify where the inverse-inference system begins to fail.

8.4 Synthetic-to-Real Evaluation

Synthetic data will form the primary supervised training source because they provide exact physical ground truth. However, the deployed system is intended to process compatible real astronomical observations.

Performance on synthetic observations will therefore not be treated as proof of real-world validity. Compatible real observations will be used as an external evaluation stage where scientifically appropriate, with the limitations of the synthetic-to-real transition explicitly documented.

The dataset is therefore designed not simply to provide a large number of images, but to provide controlled experiments linking known physical parameters to observable image structure.

9 Machine Learning System

The inverse-inference model in Olin will be implemented as a supervised deep-learning computer-vision system. Its purpose is to learn the relationship between observable lensing patterns and the selected physical parameters of the underlying lensing configuration.

The basic mapping is

$$\boxed{I_{\mathrm{obs}} \rightarrow \text{Deep Learning Model} \rightarrow \hat{\mathbf{p}}}$$

where I_{obs} is the observed image and $\hat{\mathbf{p}}$ is the predicted physical parameter vector.

9.1 Model Architecture

The primary model will be a convolutional neural network or another image-based deep-learning architecture appropriate for the final input representation.

The exact architecture will not be fixed until initial experiments establish the required input resolution, model complexity, and computational cost. The objective is to use the smallest model that provides reliable inference within the defined problem rather than maximizing architectural complexity.

A simpler machine-learning baseline may also be implemented where useful to determine whether the deep-learning model provides a meaningful improvement over simpler image-to-parameter mappings.

9.2 Training

Training will use the synthetic dataset described in Section 8, where each image has a known ground-truth parameter vector.

The model will minimize a supervised regression loss between its predictions and the known physical parameters. Training configuration, preprocessing, and dataset versions will be recorded so that model results remain reproducible.

9.3 Evaluation

Model evaluation will consider more than a single aggregate accuracy value. Performance will be measured separately for the predicted physical parameters and across controlled changes in the observation conditions.

Evaluation will include held-out synthetic observations and, where scientifically appropriate, compatible real astronomical observations.

The model will also be evaluated through the forward reconstruction described in Section 7 to determine whether predicted parameters produce physically consistent observations.

9.4 Model Artifact and Versioning

A trained model will be stored as a versioned model artifact containing the information required to reproduce its inference behaviour. This will include the trained parameters, model architecture or configuration, preprocessing requirements, output parameter definitions, and relevant training metadata.

The deployed application will load a specific validated model version rather than retraining the model during user inference.

Thus, the complete machine-learning lifecycle is

Synthetic Data → Training → Evaluation → Validated Model Artifact → Deployment

This keeps model development separate from the web application's inference workflow while maintaining a clear connection between each prediction and the model version that produced it.

10 Scientific Validation

Scientific validation in Olin will determine whether the inverse-inference model produces physically meaningful and reliable results within its defined scope. Validation will therefore evaluate both parameter prediction and consistency with the forward physical model.

10.1 Parameter Accuracy

For synthetic observations, predicted parameters will be compared directly with their known ground-truth values.

Appropriate regression metrics will be used for each predicted parameter, with errors reported both individually and collectively. Performance will also be examined across different regions of the parameter space rather than only through a single overall score.

10.2 Generalization Tests

The model will be evaluated under controlled conditions that differ from those used during training. These tests may vary physical parameters, source properties, noise levels, resolution, or other selected observational conditions.

The purpose is to determine whether the model has learned the underlying image-to-parameter relationship rather than memorizing the statistical appearance of the training distribution.

10.3 Physics-Based Reconstruction

For each suitable inference, the predicted parameters will be passed through the forward model to produce a reconstructed observation:

$$\boxed{\hat{\mathbf{p}} \rightarrow F \rightarrow \hat{I}_{\text{reconstructed}}}$$

The reconstruction will be compared with the input observation using appropriate image-level diagnostics.

This provides an independent physical consistency check: a parameter prediction that appears numerically accurate should also produce a lensing observation compatible with the original image.

10.4 Uncertainty and Failure Analysis

Where supported by the final model, Olin will report uncertainty or confidence information alongside parameter predictions.

The system will also record and analyse cases where inference becomes unreliable, particularly under conditions outside or near the boundary of the supported modelling regime.

The objective is not to hide failure cases behind an aggregate performance metric, but to identify the conditions under which the inverse model can and cannot be trusted.

10.5 Real-Observation Evaluation

Compatible real astronomical observations will be evaluated separately from the synthetic benchmark. Because complete ground-truth physical parameters are generally unavailable for real observations, real-data evaluation will not be treated as equivalent to supervised synthetic testing.

Where scientifically defensible reference estimates or established results are available, they may be used for comparison. Otherwise, evaluation will focus on physical consistency, reconstruction behaviour, and clearly stated limitations.

Scientific validation therefore follows the complete chain:

Known Parameters → Synthetic Observation → ML Prediction → Forward Reconstruction → Validation

This ensures that Olin is evaluated as a physics-grounded inference system rather than solely as an image-regression model.

11 Software Architecture

Olin will use a simple client–server architecture in which the web application communicates with a Python backend responsible for authentication, inference, report generation, and persistence.

The architecture is intentionally designed around the central scientific pipeline rather than around a large number of independent services.

11.1 System Architecture

The primary components are:

- **Frontend:** Vanilla HTML, CSS, and JavaScript for the user interface and interaction logic.
- **Backend:** Python with FastAPI for API handling, authentication, image processing, inference, and report generation.
- **Machine-Learning Model:** A validated deep-learning model loaded by the backend for image-to-parameter inference.
- **Database:** PostgreSQL for users, analysis records, predictions, model metadata, and report references.
- **File Storage:** Local filesystem storage during development and object storage for production deployment where appropriate.

The resulting high-level architecture is

$$\text{Browser} \leftrightarrow \text{FastAPI} \leftrightarrow \begin{cases} \text{ML Model} \\ \text{PostgreSQL} \\ \text{Report Storage} \end{cases}$$

The forward simulator and training pipeline are primarily offline components. The deployed backend loads a validated model artifact rather than performing model training during user requests.

11.2 Inference Workflow

A user analysis follows the sequence:

$$\begin{aligned} &\text{Upload} \rightarrow \text{Validation} \rightarrow \text{Preprocessing} \rightarrow \text{Inference} \\ &\rightarrow \text{Prediction} \rightarrow \text{Reconstruction} \rightarrow \text{Report} \rightarrow \text{Persistence} \end{aligned}$$

The backend validates that the uploaded image satisfies the supported input requirements before performing inference. Preprocessing must match the preprocessing used during model training.

The trained model then produces the selected physical parameters. These parameters are passed through the forward model to generate the physics-based reconstruction and associated diagnostics.

The prediction, reconstruction information, model version, and analysis metadata are then used to generate the scientific report.

11.3 Frontend

The frontend will use Vanilla HTML, CSS, and JavaScript.

The initial interface will provide only the functionality required for the core workflow:

1. authentication;
2. astronomical image submission;
3. analysis initiation;
4. display of inference results;
5. display of reconstruction and diagnostics;
6. scientific report generation and download;
7. access to previously generated reports.

A frontend framework such as React is not required because the initial application contains a limited number of views and relatively little persistent client-side state.

11.4 Backend

FastAPI will provide the HTTP API and coordinate the scientific inference workflow.

The backend will be responsible for:

- authentication and authorization;
- request validation;
- image validation and preprocessing;
- loading and executing the trained model;
- invoking the forward reconstruction;
- generating scientific reports;

- storing analysis metadata and report references;
- returning structured inference results to the frontend.

FastAPI does not imply that inference will automatically be distributed or asynchronous. Initial inference will be synchronous unless benchmarking demonstrates that model execution requires a different execution strategy.

11.5 Database

PostgreSQL will provide persistent relational storage.

The core relationship is

$$\text{User } 1 : N \text{ Analysis/Report.}$$

A report will be associated with the user who created it and will contain the information required to identify the analysis, predicted parameters, diagnostics, model version, and stored report artifact.

The database will store report metadata and references rather than large PDF binary data where practical.

The schema will remain intentionally small and will contain only entities required by the application's actual workflow.

11.6 Report Storage

Generated scientific reports will be persisted separately from relational metadata.

During development, reports may be stored on the local filesystem. For production deployment, object storage may be used so that the database does not become responsible for serving large binary files.

The original uploaded image will be treated as temporary processing input unless a later requirement explicitly establishes a scientific need to retain it independently.

11.7 Authentication and Authorization Boundary

Authentication will establish the identity of the user making an analysis request. Authorization will ensure that persisted analyses and reports can only be accessed by their owning user.

The scientific inference model itself will remain independent of the user's identity. Authentication exists to provide secure access to the application and persistent results rather than to alter the scientific computation.

11.8 Deployment Architecture

Olin will initially use a small number of deployable components rather than a microservice architecture.

The intended deployment is conceptually:

Web Client → FastAPI Application → PostgreSQL + Model/Report Storage

Docker will be used to provide a reproducible application environment, with CI/CD used to automate testing and deployment where practical.

Additional infrastructure such as Redis, Celery, message queues, or Kubernetes will not be introduced unless measured system requirements demonstrate a genuine need for them.

11.9 Architectural Principle

The architecture follows a central principle:

Increase the depth of the scientific pipeline, not the number of software components.

Olin is therefore intentionally designed as a contained scientific application. The software architecture should remain as simple as possible while reliably supporting the physical simulation, machine-learning inference, validation, authentication, persistence, and report-generation workflow.

12 Authentication and Authorization

Olin requires user authentication because analysis results and generated reports are persisted and must be associated with their respective users.

12.1 Authentication

Olin will support two authentication methods:

1. email and password;
2. Google OAuth.

For email and password authentication, passwords will be securely hashed and never stored in plaintext. Email verification and password-reset functionality will be provided as required by the authentication workflow.

For Google OAuth, Google will perform the identity authentication. Olin will associate the authenticated Google identity with a local application user record. Google credentials will not be stored or handled by Olin.

Authenticated sessions will use an access-token and refresh-token mechanism. Authentication endpoints will include appropriate rate limiting to reduce brute-force, credential-abuse, and verification-code abuse.

12.2 Authorization

Authorization will be ownership-based rather than role-based.

Each persisted analysis or report will be associated with a specific user. Access will be permitted only when the authenticated user's identity matches the owner of the requested resource.

Conceptually,

$$\boxed{\text{current_user.id} = \text{resource.user_id}}$$

Olin will not implement roles, administrative permissions, organizations, or multi-tenant access controls in V1.

12.3 Credential and OAuth Isolation

Olin will maintain its own authentication configuration and OAuth credentials. Existing authentication patterns or implementation knowledge from other projects may be reused where appropriate, but production credentials, secrets, and application-specific configuration will remain isolated.

Authentication is therefore treated as a supporting application subsystem. It must protect persisted analyses without becoming part of the scientific inference pipeline.

13 Deployment and Operations

Olin will use a lightweight deployment strategy focused on reproducibility, maintainability, and sufficient resources for the initial inference workload.

13.1 Containerization

Docker will be used to package the application and its runtime dependencies. This will provide a consistent environment across development, testing, and deployment.

The deployment will remain small and will not require a microservice architecture.

13.2 Compute Strategy

The initial deployment will be CPU-first. Model inference performance will be measured before introducing GPU infrastructure.

GPU deployment will only be considered if the validated model demonstrates a measured computational requirement that materially affects the user workflow.

13.3 Continuous Integration and Deployment

A CI/CD pipeline will be used to automate appropriate checks such as:

- application tests;
- backend validation;
- dependency and build checks;
- container image construction;
- deployment of validated application versions.

Model artifacts will be versioned independently from application code where appropriate, while each analysis will record the model version used for inference.

13.4 Storage and Configuration

Application secrets, database credentials, OAuth credentials, and other environment-specific configuration will be supplied through deployment configuration rather than committed to source control.

Generated reports will use the storage architecture defined in Section 11: local filesystem storage during development and appropriate object storage for production where required.

13.5 Operational Simplicity

Olin will not introduce Redis, Celery, Kubernetes, message queues, or other distributed infrastructure by default.

If inference latency, concurrency, storage volume, or another measured requirement exceeds the capabilities of the initial architecture, the relevant component may be introduced through an explicit architectural revision.

The operational principle is therefore:

Measure first → identify the actual bottleneck → add infrastructure only when justified

14 Architecture Decision Records

The following architectural decisions define the intended V1 structure of Olin. Each decision is based on the requirements of the scientific pipeline rather than familiarity with a particular technology.

14.1 ADR-01: FastAPI Backend

Decision: Use Python with FastAPI for the backend.

Reason: The core machine-learning and scientific-computing components are implemented in Python. Keeping the API and inference system within the same runtime avoids an unnecessary language and service boundary.

Spring Boot, Django, Flask, and other backend frameworks remain technically possible, but do not provide a sufficiently strong requirement-based reason to introduce additional complexity.

14.2 ADR-02: Vanilla JavaScript Frontend

Decision: Use HTML, CSS, and Vanilla JavaScript.

Reason: Olin has a small number of views and limited client-side state. A frontend framework would introduce additional build and dependency complexity without solving a significant problem in the initial application.

14.3 ADR-03: PostgreSQL Database

Decision: Use PostgreSQL for persistent application data.

Reason: Users, analyses, reports, model metadata, and ownership relationships form a small relational data model. PostgreSQL provides transactions, constraints, and reliable relational queries appropriate for this workflow.

MongoDB is not selected because the application does not have a strong document-oriented requirement that would justify replacing the relational model.

14.4 ADR-04: Monolithic Application

Decision: Use a single primary backend application rather than microservices.

Reason: Olin has a small and well-defined workflow. Separating authentication, inference, reporting, and persistence into independent services would increase deployment and operational complexity without a demonstrated benefit.

14.5 ADR-05: No RAG or LLM

Decision: Do not include retrieval-augmented generation or a large language model in the V1 inference pipeline.

Reason: Olin performs structured physical inference rather than document retrieval or natural-language generation. Its reports can be generated deterministically from structured predictions, diagnostics, and metadata.

Adding an LLM would introduce an additional probabilistic subsystem without solving a required problem.

14.6 ADR-06: Controlled Synthetic Training Data

Decision: Use a physics-based synthetic dataset as the primary supervised training source.

Reason: Synthetic observations provide exact ground-truth physical parameters and allow controlled variation of physical and observational conditions. Real observations generally do not provide equivalent complete ground truth.

Real astronomical observations will therefore be treated as a separate evaluation stage rather than automatically becoming training data.

14.7 ADR-07: CPU-First Deployment

Decision: Begin deployment using CPU inference.

Reason: Hardware complexity should be driven by measured model requirements. GPU infrastructure will only be introduced if inference benchmarks demonstrate a meaningful need.

14.8 ADR-08: Physics-Based Reconstruction

Decision: Pass predicted parameters through the forward model after inference.

Reason: Reconstruction provides a physical consistency check in addition to numerical parameter accuracy. It also connects the deployed machine-learning model back to the physical assumptions used to generate its training data.

14.9 ADR-09: Minimal Application Scope

Decision: Prioritize the central inference workflow over additional application features.

Reason: The scientific value of Olin lies primarily in the physics–simulation–ML–validation pipeline. Additional dashboards, analytics, social features, complex user profiles, or distributed infrastructure would increase implementation cost without strengthening the central scientific objective.

These decisions may be revised only when a demonstrated scientific, performance, or operational requirement provides a clear reason to do so.

15 Functional Requirements

The following requirements define the core behaviour that Olin V1 must provide. A requirement is considered satisfied only when the corresponding behaviour can be demonstrated and tested.

15.1 User and Authentication Requirements

- **FR-01:** The system shall allow users to create accounts using email and password.

- **FR-02:** The system shall support Google OAuth authentication.
- **FR-03:** The system shall authenticate users before accessing protected analysis and report resources.
- **FR-04:** The system shall provide password recovery for email-based accounts.
- **FR-05:** The system shall allow authenticated users to access only their own persisted analyses and reports.

15.2 Image Analysis Requirements

- **FR-06:** The system shall allow an authenticated user to submit a compatible astronomical image for analysis.
- **FR-07:** The system shall validate the submitted image before inference.
- **FR-08:** The system shall apply the preprocessing procedure required by the deployed model.
- **FR-09:** The system shall execute the validated deep-learning model on the submitted observation.
- **FR-10:** The system shall return predictions for the physical parameters defined by the active model version.
- **FR-11:** The system shall generate a physics-based reconstruction using the predicted parameters.

15.3 Report Requirements

- **FR-12:** The system shall generate a scientific report for a completed analysis.
- **FR-13:** The report shall contain the predicted physical parameters and relevant diagnostics.
- **FR-14:** The report shall identify the model version used for the analysis.
- **FR-15:** The report shall include the physics-based reconstruction produced from the inferred parameters.
- **FR-16:** The system shall allow the authenticated user to retrieve and download reports belonging to that user.

15.4 Persistence Requirements

- **FR-17:** The system shall persist completed analysis metadata and report references.
- **FR-18:** Each persisted analysis shall be associated with the user who initiated it.
- **FR-19:** Each completed analysis shall record the model version used to produce its prediction.

15.5 Model and Reproducibility Requirements

- **FR-20:** The deployed application shall load a validated model artifact rather than train a model during user inference.
- **FR-21:** The system shall preserve the preprocessing and output configuration required by the active model version.
- **FR-22:** The system shall retain sufficient metadata to associate an analysis result with its model and analysis configuration.

15.6 Scientific Workflow Requirements

- **FR-23:** The synthetic-data pipeline shall generate observations from the defined forward physical model.
- **FR-24:** Each synthetic observation used for supervised training or evaluation shall have known ground-truth parameters.
- **FR-25:** The scientific evaluation pipeline shall measure model performance on held-out observations.
- **FR-26:** The evaluation pipeline shall support controlled generalization experiments within the defined physical regime.
- **FR-27:** The system shall keep synthetic-data training results distinct from evaluation of compatible real observations.

These requirements define the minimum functional behaviour of Olin V1. Features outside this set shall not be treated as implicit requirements.

16 Non-Functional Requirements

The following requirements define the quality, security, reliability, and maintainability properties expected from Olin V1.

16.1 Performance

- **NFR-01:** The system shall provide inference performance suitable for interactive use under the deployed hardware configuration.
- **NFR-02:** Model inference performance shall be benchmarked before introducing additional compute infrastructure.
- **NFR-03:** Image validation and preprocessing shall not introduce unnecessary processing overhead.

16.2 Security

- **NFR-04:** Passwords shall never be stored in plaintext.
- **NFR-05:** Protected resources shall require valid authentication.
- **NFR-06:** Users shall not be able to access analyses or reports belonging to other users.
- **NFR-07:** Authentication and resource-intensive endpoints shall implement appropriate rate limiting.
- **NFR-08:** Uploaded files shall be validated for supported type and reasonable size before processing.
- **NFR-09:** Application secrets and OAuth credentials shall not be committed to source control.
- **NFR-10:** Production communication shall use HTTPS.

16.3 Reliability

- **NFR-11:** Invalid or unsupported inputs shall produce controlled errors rather than uncontrolled application failures.
- **NFR-12:** A failed analysis shall not produce a report that appears to represent a successful inference.
- **NFR-13:** Persisted analysis records shall remain associated with the correct user and model version.

16.4 Reproducibility

- **NFR-14:** Model artifacts shall be versioned.
- **NFR-15:** Dataset-generation and model-training configurations shall be recorded for scientific experiments.
- **NFR-16:** An analysis shall identify the model version and relevant configuration used to produce its result.

16.5 Maintainability

- **NFR-17:** The application shall maintain a clear separation between frontend, API, scientific inference, persistence, and report-generation responsibilities.
- **NFR-18:** The system shall use automated testing and version-controlled source code.
- **NFR-19:** Deployment shall be reproducible through the defined containerized environment.

16.6 Usability

- **NFR-20:** The primary analysis workflow shall be accessible without requiring users to install scientific or machine-learning software locally.
- **NFR-21:** Inference results shall be presented together with sufficient context to distinguish predictions, reconstructions, and diagnostic information.

These requirements define the expected quality of the system without imposing infrastructure or complexity that is not justified by the application's actual workload.

17 Final System Definition

Olin V1 is an authenticated web-based scientific application for gravitational-lensing inverse inference.

The system will use an established, parameterized gravitational-lensing forward model to generate controlled synthetic observations with known physical parameters. These observations will be used to train and evaluate a supervised deep-learning computer-vision model that estimates selected physical parameters from compatible astronomical images.

The deployed application will accept compatible real astronomical observations, preprocess them according to the active model version, perform inference, and return the predicted physical parameters. The predictions will then be passed through the forward model to produce a physics-based reconstruction of the observation.

Each completed analysis will produce a persistent scientific report containing the relevant predictions, reconstruction, diagnostics, model version, and analysis metadata. Authentication and ownership controls will ensure that persisted analyses are associated with the correct users.

The complete V1 workflow is therefore:

$$\boxed{\text{Physical Model} \rightarrow \text{Synthetic Data} \rightarrow \text{DL Training}}$$

Validation → Real-Image Inference

Reconstruction → Scientific Report

The application will use a deliberately contained software architecture consisting of a web frontend, FastAPI backend, PostgreSQL database, trained model artifact, and report storage. Additional infrastructure or scientific complexity will only be introduced when justified by measured requirements or validated future extensions.

Olin V1 is therefore defined not as a general astronomical analysis platform, but as a focused physics-grounded inverse-inference system connecting gravitational-lensing simulation, deep learning, scientific validation, and accessible software delivery.